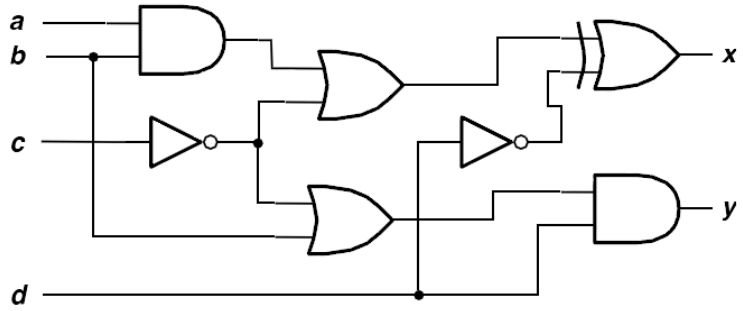
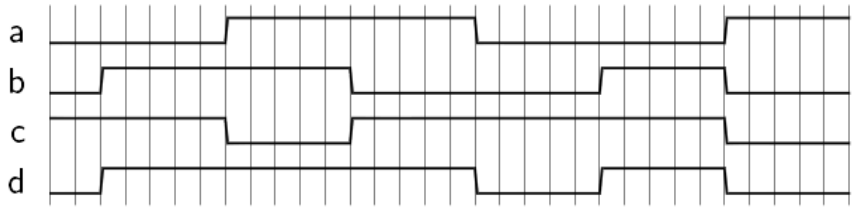
CE2003 Tut 5

Q5 (a) For the following circuit, determine the propagation delay from each input to each output. Assume a delay for an **and** gate is 3ns, an **or** gate is 2ns, an **inverter** is 1ns, and an **xor** gate is 4 ns:



(b) Hence, what is the critical path? How fast could such a circuit be clocked if registers were placed at the inputs and outputs?

(c) How would the circuit react to the following inputs (assume each vertical gridline is 2ns)? Show the outputs at each gate before working out the overall output. Can you identify any glitches?



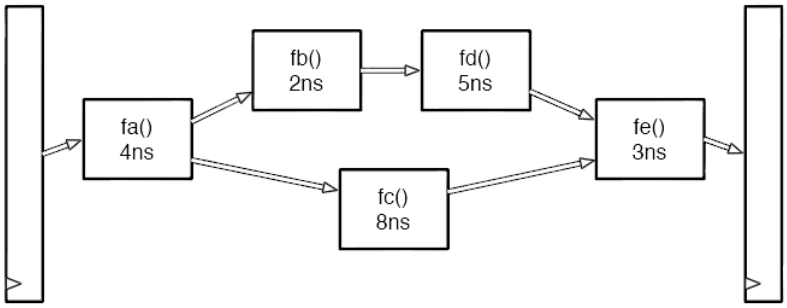
Q2 (a) Assuming the propagation delay through an **and** gate or **or** gate is equal to the number of inputs plus 1ns, determine the propagation delay through a minimised sum-of-products implementation of the function $\sum \mathrm{m}(0, 5, 8, 10, 11, 13, 14, 15)$

(b) How would this be implemented in an FPGA, and what would be the propagation delay?

(c) Assuming the above gate-level circuit was placed between two register stages with a flip flop clock-to-Q delay of 0.6ns, and a setup time of 0.3ns, how fast could the circuit be clocked?

(d) You now place an extra set of flip flops, with the same characteristics between the **and** outputs and the **or** input. How fast can the circuit be clocked now?

Q3 You are designing a circuit with multiple combinational functional blocks as shown in the diagram. Each block shows its function name, and its propagation delay.



(a) How fast can the circuit be clocked as is?

(b) What would be the maximum frequency achievable assuming we cannot pipeline the contents of the functional blocks?

(c) Draw a pipelined implementation that achieves this maximum speed, with a minimum required number of extra pipeline stages added.

(d) Assume the following sequence of samples are fed into the system at successive clock edges, draw a timing diagram to show the outputs at each register stage:

s1, s2, s3, s4, s5, s6, s7, s8, s9, s10